CHAEWON YOON

Madison, WI | **Email** | **GitHub** | **LinkedIn** | **Hugging Face** | **ORCID**

RESEARCH FOCUS

Independent AI researcher working on compute-efficient inference, numerical precision, neural operators, and causal evaluations of model behavior.

EDUCATION

University of Wisconsin-Madison | B.S. Data Science (in progress) *May 2028*

Transferred from University of Wisconsin Oshkosh (Fall 2026); prior independent studies in numerical PDEs, neural operators, and stochastic systems.

SELECTED ACCEPTED PUBLICATIONS

- **WACV 2027, Algorithms Track (accepted).** Yoon, C. *The Measurement Operator Changes Which Bits Matter: Operator-Conditioned Mixed Precision for Inverse Vision.* **Paper**
- **ACM UIST Adjunct 2026 (poster).** Yoon, C. *EvidenceLens: A Declarative Evidence-State Interface for Human-AI Visual Inspection.* **DOI**

SELECTED RESEARCH EXPERIENCE

Operator-Conditioned Mixed Precision | Independent Research *2026*

- Designed Operator-Susceptibility Matching to allocate three high-precision stages in an eight-stage inverse-vision model; across 256 candidate precision schedules, reduced mean operator-visible regret to **2.7% versus 6.3%** for the best global-static schedule.
- In a separate native A100 timing evaluation, the selected mixed-precision configuration took **16.4 ms versus 23.6 ms** per reconstruction for an FP16 reference (**1.44x speedup**). Quality and timing use distinct protocols. **Study**

Quantized Defect Correction | Faculty-Mentored Scientific ML *2025-2026*

- Built reproducible PyTorch Fourier neural operator workflows for fractional PDEs, including Q6 activation/weight sensitivity tests, disjoint out-of-distribution holdouts, physics-residual diagnostics, and matrix-free LOBPCG correction. **Artifact**

Multilingual LLM Mechanisms | Collaborative Research *2026*

- Coauthored an **ACL Rolling Review manuscript (under review)** testing single-head language sensitivity across 12 models from seven series; seven heads shifted 8.0%-51.5% of FLORES continuations off-language, versus at most 1.4% for other same-layer heads.
- Contributed merged Qwen/Llama circuit discovery and activation-patching studies with held-out FLORES tests and 100 layer-matched random-head controls. **Qwen PR** | **Llama PR**

SELECTED ENGINEERING AND EVALUATION CONTRIBUTIONS

- Merged AladdinAI fixes for deterministic 70/15/15 train/validation/held-out splits, tool-error recovery tests, DOCX parsing, and shared frontend error handling. **Data split** | **Frontend**
- Built and merged Station House OLMo 3 identity-spoofing evaluation with 9 experimental conditions, 27 conversations, and 3 scoring approaches. **Merged PR**

FELLOWSHIP AND TEACHING

WAISI Technical AI Safety Fellow, Oct. 2026-present (weekly technical reading program); **UW-Madison PEOPLE math mentor**, Oct. 2026-present; **UW Oshkosh mathematics tutor**, Sep. 2025-May 2026 (10-15 students weekly).

TECHNICAL SKILLS

Python, PyTorch, NumPy, SciPy, TensorFlow, Git, LaTeX; operator learning, mixed precision, numerical PDEs, activation patching, held-out evaluation, regression testing.